

Absorption current and depolarisation measurements on polypropylene film

This content has been downloaded from IOPscience. Please scroll down to see the full text.

1988 J. Phys. D: Appl. Phys. 21 454

(<http://iopscience.iop.org/0022-3727/21/3/013>)

View [the table of contents for this issue](#), or go to the [journal homepage](#) for more

Download details:

IP Address: 134.129.120.3

This content was downloaded on 24/05/2015 at 10:27

Please note that [terms and conditions apply](#).

Absorption current and depolarisation measurements on polypropylene film

M Tavakoli† and J Hirsch

Birkbeck College (University of London), London WC1E 7HX, UK

Received 15 July 1987, in final form 29 September 1987

Abstract. We report absorption and resorption current measurements on low-loss polypropylene over ranges of 10 μ s to 25 h, 0.5 to 2 MV cm⁻¹, and -175 to 100 °C. While the resorption transients always decay hyperbolically, above 20 °C the absorption transients develop a conduction tail which is insensitive to contact metal. The current is resolved into volume and surface components, whose variation with time, field and temperature is illustrated. TSD measurements show no significant charge injection from Al contacts, but electron-beam-induced depolarisation experiments disclose heterocharge. Very deep electron traps (depth $\geq$ 0.9 eV) and, tentatively, hole traps $\approx$ 0.75 eV deep are identified by TSD on electron-bombarded films. All the observations point to an ionic mechanism, with a threshold related to glass transition, for the conduction tail.

1. Introduction

The absorption or resorption current in a sample capacitor is defined as the transient current which continues to flow in response to a voltage transition, after the decay of the displacement current due to the geometric capacitance and electronic polarisation. In previous work in this laboratory, high-field transient measurements were used with good effect to clarify the role of contact-injected electronic carriers in polyethylene terephthalate (PET) (Ko and Hirsch 1981, Ko 1984, Hirsch *et al* 1984). The conclusions drawn were confirmed by complementary measurements of thermally stimulated and electron-bombardment-induced depolarisation (TSD and EBD) on samples which had been either voltage polarised or implanted with electrons under prior strong bombardment.

In PET, we found that the 'conduction tail' of the absorption transient was sensitive to the electrode metal. A substantial charge of electrons, strongly increasing with field and temperature, was injected by an Al cathode, and we associated the conduction current with this charge‡. Conduction tails are observed at high temperatures and fields in most polymers. In polypropylene (PP), they were first studied by Foss

and Dannhauser (1963), who attributed them to ionic conductivity. Das Gupta and Joyner (1976) investigated the effects of different electrode metals on the transients in PP, but found no differences. These authors suggested a bulk electronic mechanism for the tail.

We undertook a further study of PP in the hope that the diagnostic techniques, previously applied to PET, would clarify the origin of conduction in PP also. In PET, we had confirmed that the transient current was predominantly of bulk character. In PP, which possesses much lower dielectric loss, we suspected a significant contribution from surface leakage (Lachish and Steinberger 1974). This tends to grow in prominence the lower the loss (Atkinson and Fleming 1980a), but had not been considered by the earlier workers. A brief account of the experiments described below has already been published (Tavakoli and Hirsch 1987). Full details of the apparatus, methods and results are given in the thesis by Tavakoli-Juibari (1987).

2. Experimental details

The samples were cut from 8 μ m thick ICI Propafilm 'E', nominally additive-free and kindly donated by Professor W Reddish, lately of ICI plc. Its density§ corresponded to a crystallinity of $\approx$ 65% (Quynn *et al* 1959). The mean molecular mass was $\approx$ 10⁵, the glass transition temperature $\approx$ -8 °C, and the onset of melting in the crystalline phase $\approx$ 140 °C. The surfaces were ultrasonically cleaned in isopropyl alcohol

† Present address: Department of Electrical Engineering and Electronics, University of Liverpool, Liverpool L69 3BX, UK.

‡ Recently, however, Kurtz and Anderson (1986) concluded that conduction in PET could not be of electronic but must be of ionic origin.

§ 0.915 g cm⁻³, measured by flotation in an isopropyl alcohol/water mixture. In Tavakoli and Hirsch (1987) the crystallinity was given in error as 80–90%. Other determinations by gel permeation chromatography and DSC by RAPRA Technology Ltd.

before vacuum depositing circular electrodes of Al or Au.

The live electrode was always of 20 mm diameter. For the transient current experiments the counter electrode, which was connected to the virtual-ground input of the detector, was of variable diameter (see § 3.2), but typically 14 mm. It was surrounded by a concentric, earthed, guard ring of 17 mm internal and 20 mm external diameter to suppress tracking around the edge of the film. For the depolarisation experiments, the counter electrode was of 20 mm diameter and unguarded, and was connected to an electrometer.

The apparatus for the transient measurements (see figure 1) has already been described (Hirsch *et al* 1984, Ko 1984), but the sample holder and electronic circuits were redesigned and improved for the present project. The saturated-transistor switch provided voltage steps up to 1.6 kV (corresponding to an average field of 2 MV cm^{-1}) with rise or fall times $\leq 1 \mu\text{s}$. The detector was a low-leakage, low-drift operational integrator. Its input was shorted while idling by forward-biased, current-stabilised, pA diodes in a self-zeroing configuration. The short circuit was maintained for $4 \mu\text{s}$ after the high-voltage transition to bypass the initial displacement current (typically 0.5 A). Internal switching surges were compensated by the infinite-track/hold arrangement shown, which had a conversion time of $\leq 1 \mu\text{s}$.

Although a dead time of $5 \mu\text{s}$ was adequate, a first charge sample was normally taken only after $10 \mu\text{s}$, with subsequent samples at 20, 40, 80... μs . All were stored in a transient recorder, and the current was later computed by logarithmic differentiation. The resolution was 1 pA, but when the current had decayed to approximately 10 pA (typically after 2 s), it was switched into an electrometer to extend the record.

The measurements were carried out in high vacuum. The temperature was varied between -175 and 100°C , and the field between 0.5 and 2 MV cm^{-1} . The voltage polarity was regularly reversed to check for asymmetries. Prior to any absorption measurement, the sample was annealed in

short circuit at approximately 115°C for 30 min to discharge any remnant polarisation.

The procedure for TSD was standard. Ramp heating was carried out under dry, oxygen-free, nitrogen gas at atmospheric pressure to limit the temperature difference between the faces of the sample to $\leq 2^\circ\text{C}$. The EBD method used was that previously described by Ko and Hirsch (1981) and Ko (1984). In it, a polarised sample is subjected to repetitive bombardment by weak electron pulses. The conductivity temporarily induced gives rise to discharge transients driven by the internal polarisation field. The sign of these transients depends on that of the internal field, and makes it possible to distinguish between homocharge (injected charge) and heterocharge (as expected in ionic polarisation).

3. Experimental results

3.1. Transient measurements

The representative absorption (I_a) and resorption (I_r) transients in figure 2 refer to aluminised samples with 14 mm diameter, guarded, counter electrodes. At all temperatures (T) and average fields (F), $I_r(t)$ decays as t^{-n} where $n = 1(+0-2\%)$ with great uniformity from $10 \mu\text{s}$ to 15 min or longer. At and below room temperature (RT), $I_a(t)$ is an exact mirror image of $I_r(t)$, and is not displayed separately in the diagram. At 100°C , I_a mirrors I_r initially, but later develops a conduction tail. The profile of the tail is almost invariant with F . The current is still in decay after 25 h (curve A), and shows no tendency to level out. Transients in samples with two electrodes of Au, or one of Al and one of Au irrespective of polarity, are indistinguishable from those illustrated for two Al electrodes at all F and T , i.e. there is no resolvable electrode effect.

In the isochronal current-temperature 'spectra' of figure 3, the curve in (b) for $I_r(T)$ at 100 s replicates the shape of the curve in (a) for $I_a(T)$ or $I_r(T)$ at 0.1 ms on account of the uniform decay of $I_r(t)$ at all T . The broad, weak, peak at approximately -70°C

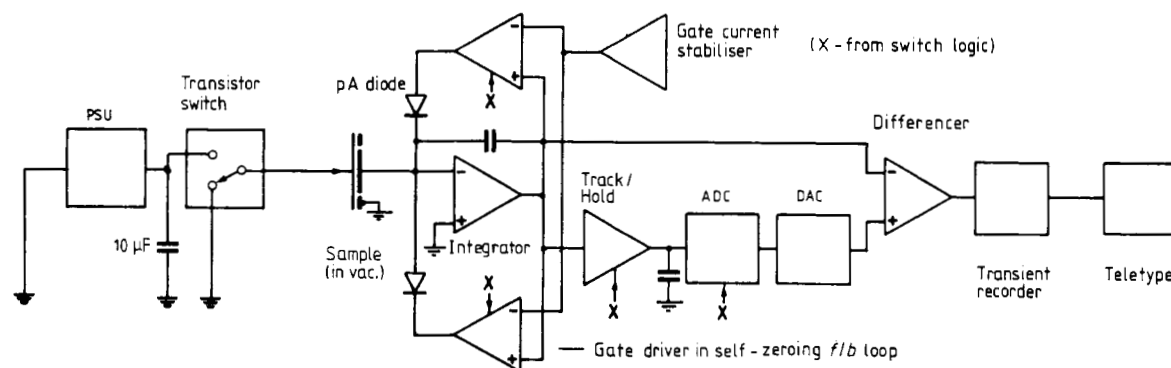


Figure 1. A block diagram of the arrangement for transient current measurements.

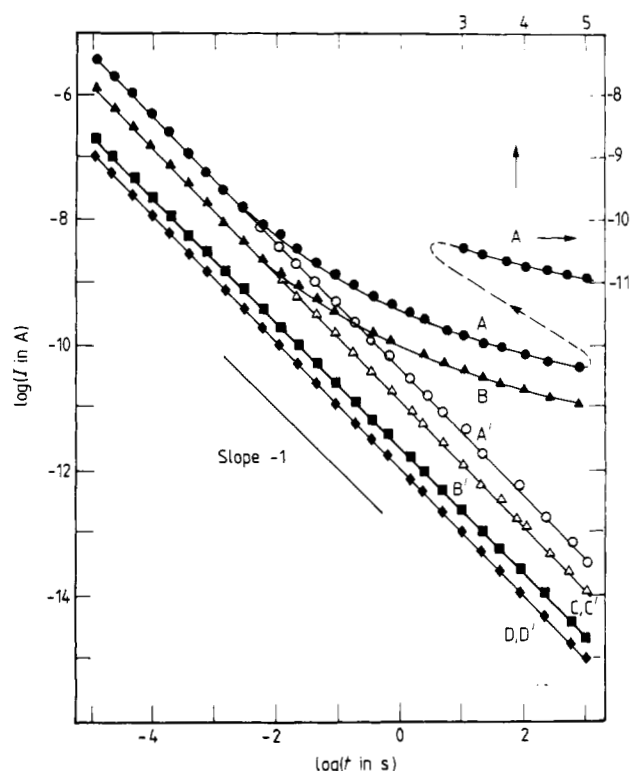


Figure 2. Absorption and resorption current transients (resorption marked by primed letters). A, A': 100 °C, 2 MV cm⁻¹; B, B': 100 °C, 0.5 MV cm⁻¹; C, C' (coincident): 20 °C, 2 MV cm⁻¹, ÷ 10; D, D' (coincident): -175 °C, 2 MV cm⁻¹, ÷ 10. Al electrodes, live 20 mm diameter, counter 14 mm diameter.

does not appear to shift with sample time. The rise of $I_a(T)$ at 100 s only just above RT (figure 3(b)) reflects the tail current. An Arrhenius plot of this rise above I_r yields an activation energy of approximately 0.8 eV.

3.2. Surface leakage

We wish to make it clear that the objective of this investigation was not a general study of surface leakage, but an attempt to resolve volume from surface currents under the particular specimen geometry employed. For this purpose, the transient measurements were repeated on aluminised samples with counter electrodes of diameter $d = 5, 7$, and 10 mm, without changing the guard ring dimensions (§ 2).

At and below RT, $I_a(t)$ and $I_r(t)$ are exact mirror images of each other, and decay hyperbolically irrespective of diameter. However, for example at RT and 1.5 MV cm⁻¹, the current levels for diameters of 14 and 5 mm are in a ratio of ≈ 4 , whereas the area ratio is ≈ 8 . At the same field but at 100 °C, the I_r transients for the same two diameters again differ by a factor ≈ 4 , but the tail currents at 10³ s differ by ≈ 8 . This points to the presence of a time-dependent surface component. Polarity reversal had no resolvable effect.

Following Atkinson and Fleming (1980a), we

assumed that the total measured current I_a or I_r could be expressed as the sum of a volume (or bulk) component I_v proportional to d^2 , and a surface (or perimeter) component I_s proportional to d . That is

$$I_a = I_{av} + I_{as} = \alpha_a(t, F, T)d^2 + \beta_a(t, F, T)d \quad (1)$$

in absorption and, similarly, with subscript r , in resorption. Plots of I_a/d and I_r/d for sets of selected values of t , F and T yielded acceptable straight lines, from which α and β could be deduced. Since many separate experiments on different samples were involved, the accuracy of these determinations is difficult to estimate, but I_v/I and I_s/I generally added up to $1 \pm 5\%$.

We illustrate these results in some detail, since they may be of general interest, and such curves have not been published previously for PP or any other polymer. In figures 4–7, $v = I_v/I$ represents the volume fraction and $s = I_s/I$ the surface fraction of the total current, for $d = 14$ mm. Plots for different d may be constructed from the curves by the use of equation (1). Figures 4(a) (for RT) and 4(b) (for 100 °C) show the evolution of v_a and s_a in absorption, and figures 5(a) (RT) and 5(b) (100 °C) the field dependence of v_a and s_a in the hyperbolic decay region (0.1 ms) and tail region (100 s) of $I_a(t)$. At RT, v_r and s_r are identical with v_a and s_a (figures 4(a) and 5(a)), since $I_a(t)$ and $I_r(t)$ coincide (cf figure 2). In contrast, at 100 °C the magnitudes of v_r and s_r (figure 6(a)) and their field dependences (figure 6(b)) are seen to be the reverse of those in figures 4(b) and

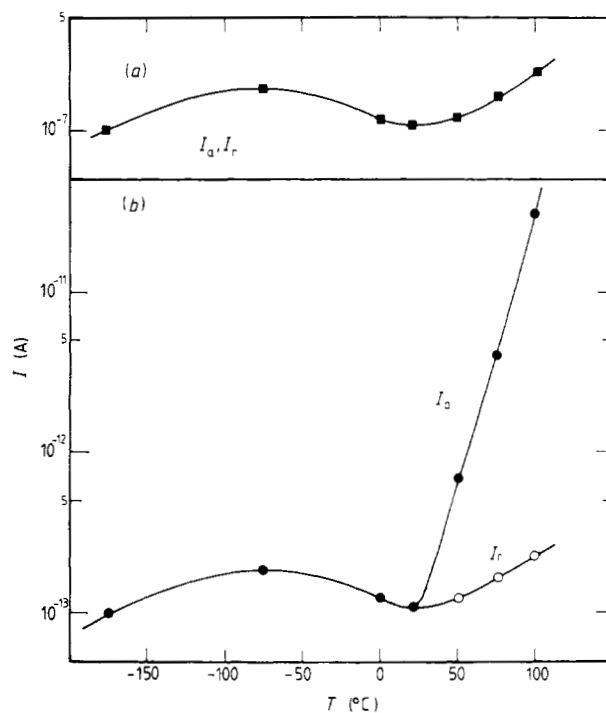


Figure 3. Isochronal current-temperature spectra in (a) the initial decay region (0.1 ms) and (b) the tail region (100 s). I_a , absorption; I_r , resorption current. Electrodes as in figure 2.

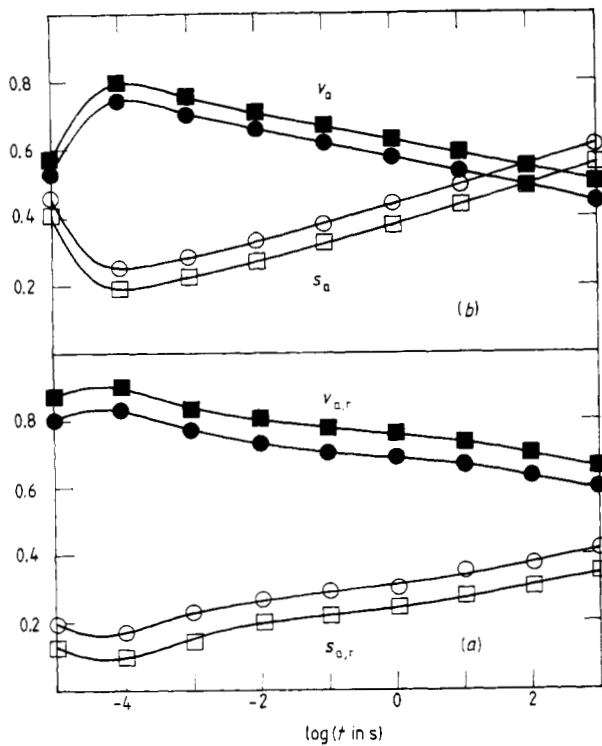


Figure 4. The evolution of volume and surface current fractions v and s . (a) 20 °C, absorption or resorption (coincident); (b) 100 °C, absorption. ■ v , □ s , 1.5 MV cm⁻¹; ● v , ○ s , 0.5 MV cm⁻¹. Electrodes as in figure 2.

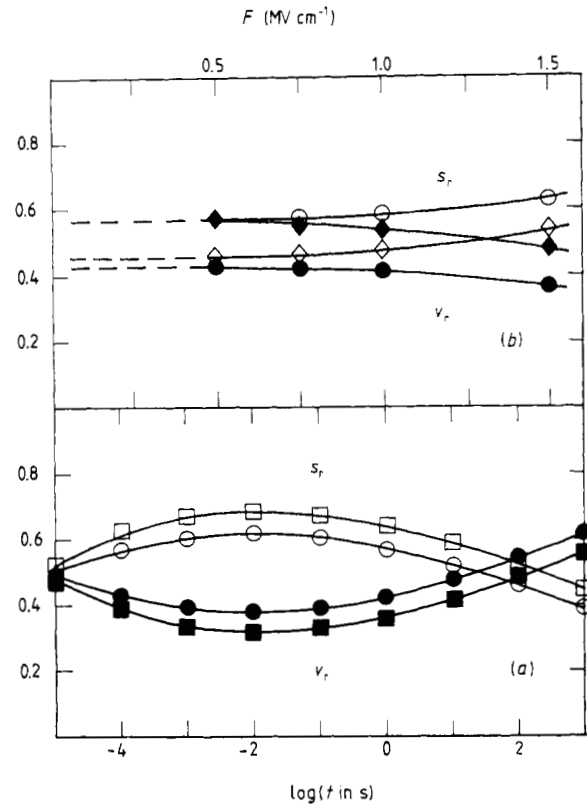


Figure 6. The resorption at 100 °C. (a) Evolution of volume and surface current fractions v and s . ■ v , □ s , 1.5 MV cm⁻¹; ● v , ○ s , 0.5 MV cm⁻¹. (b) The field dependence of v and s . ● v , ○ s , 0.1 ms; ◆ v , ◇ s , 100 s. Electrodes as in figure 2.

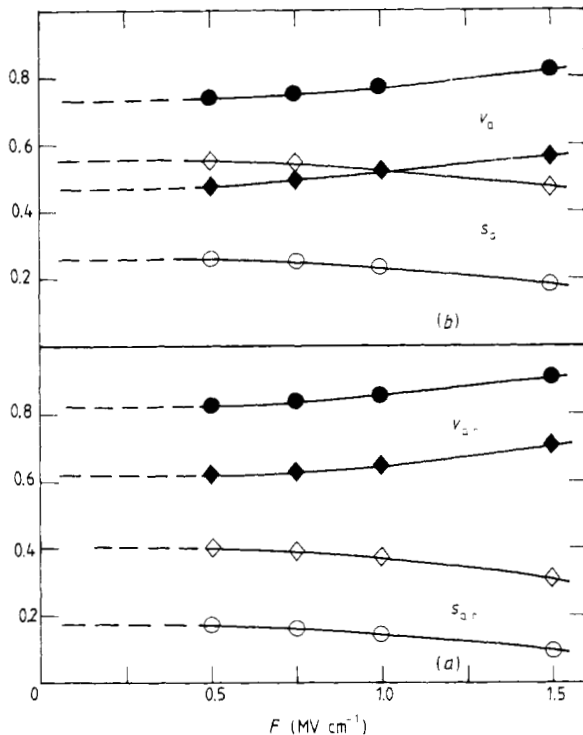


Figure 5. The field dependence of the isochronal volume and surface current fractions v and s . (a) 20 °C, absorption or resorption (coincident), (b) 100 °C, absorption. ● v , ○ s , 0.1 ms; ◆ v , ◇ s , 100 s. Electrodes as in figure 2.

5(b). The complexity of the results is emphasised by plotting the evolution of the ratio s/v , especially at 100 °C (see figure 7).

Figures 5 and 6(b) suggest that v and s may become independent of field as F tends to zero, as indicated by the broken curve extrapolations. The zero-field values of s so deduced are listed in table 1, together with a parameter $c = (v/s)/1.4$ where v and s are for a 14 mm diameter electrode. By means of this parameter, the surface current fraction $s(d)$ at $t = 0.1$ ms or 100 s, $T = 20$ or 100 °C, and $F = 0$ may be calculated for any given diameter d in cm, using the relation

$$s(d) = (1 + cd)^{-1}. \quad (2)$$

3.3. Field dependence of isochronal current

Corrections for surface current have a negligible effect on the patterns of figures 2 and 3, but are vital to the interpretation of the field dependence of the isochronal current. We illustrate this in figure 8 by two examples, namely the field characteristics of I_a (the total measured absorption current) and its resolved components I_{av} and I_{as} at 100 °C, at the two sample times previously chosen to represent the hyperbolic decay and tail regions of $I_a(t)$. The results

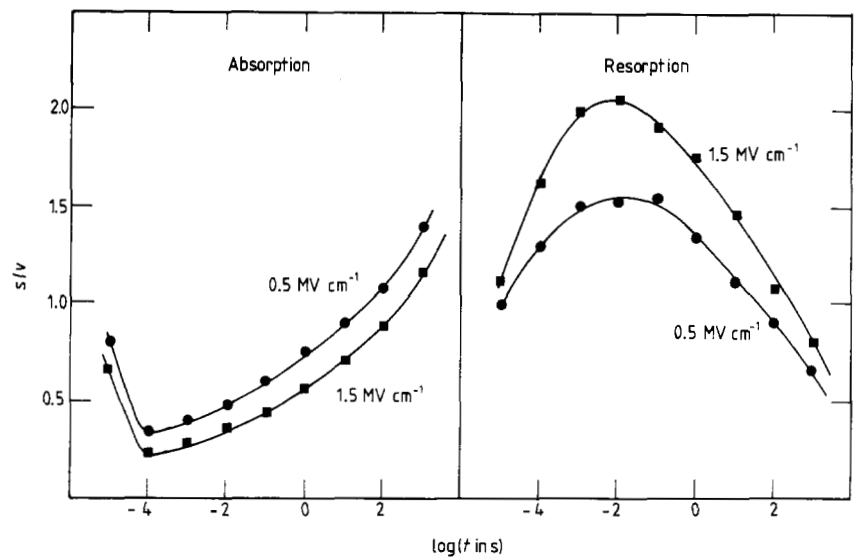


Figure 7. The evolution of the surface/volume current ratio s/v at 100 °C. Electrodes as in figure 2.

of these and other plots may be summarised as follows: the field dependences of I_a and I_r generally are non-ohmic; those of I_{as} and I_{rs} are always strongly non-ohmic. However, at 0.1 ms, I_{av} and I_{rv} are strictly ohmic in all cases and, at 100 s, I_{av} is only weakly supraohmic, varying as $\sim F^{1.1}$. I_r at 100 s is so low and the fraction I_{rs}/I_r so large that the field dependence of I_{rv} cannot be reliably determined, but it is approximately ohmic or at most slightly supraohmic.

3.4. TSD experiments

Curve A in figure 9 represents a typical short-circuit TSD ‘thermogram’ for an aluminised specimen voltage polarised at 80 °C. The prominent peak at $T_m \approx 115$ °C corresponds to an activation energy $E \approx 0.85$ eV, as deduced both from initial rise analysis and the method of Grossweiner (1953). The weak peak at approximately 35 °C corresponds to $E \approx 0.65$ eV by the $25 kT_m$ rule (Milnes 1973). Both peaks increase in height with increasing polarising time, field and temperature. The field dependence of the height of the 115 °C peak is supraohmic ($\sim F^2$), but this result is unreliable as it may have been distorted by surface leakage.

Table 1. Low-field (extrapolated) values of surface current fraction s for a 14 mm diameter Al electrode in absorption (a) or resorption (r), and parameter c (per cm diameter) in equation (2).

t (s)	20 °C		100 °C			
	$s_{a,r}$	$c_{a,r}$	s_a	c_a	s_r	c_r
10^{-4}	0.18	3.3	0.27	1.9	0.55	0.6
10^2	0.4	1.1	0.45	0.9	0.45	0.9

Curve B in figure 9 is a short-circuit thermogram taken after strong electron bombardment at RT in short circuit. The differences between curves B and A are striking. In curve B, the TSD current is finite at 0 °C, and increases with temperature in the form of a broad shoulder leading up to a main peak at approximately 130 °C, close to the softening temperature of the polymer. The resolution of the experiment is sufficient for us to be confident that the main peaks in curves A and B do not correspond to each other. That is, the 115 °C peak of A is absent in B. The 35 °C peak of A is just discernible in B, and there is a new minor peak at 80–90 °C. Short-circuit thermo-

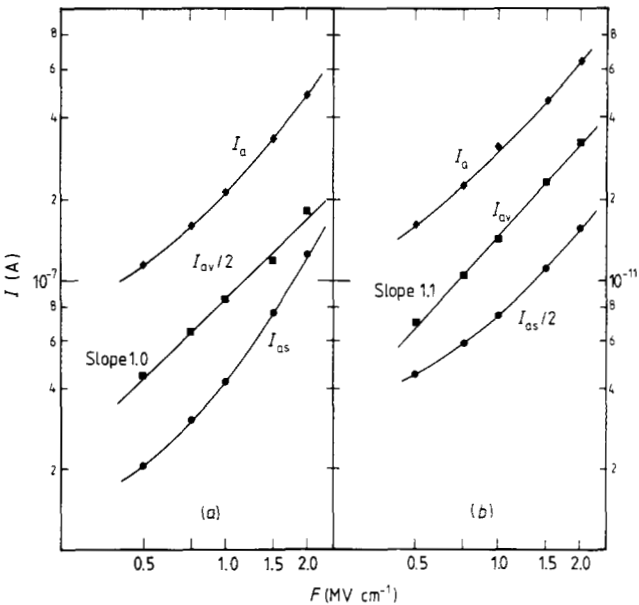


Figure 8. The field dependence of the total absorption current I_a and its resolved volume and surface components I_{av} and I_{as} at 100 °C and (a) 0.1 ms, (b) 100 s. Electrodes as in figure 2.

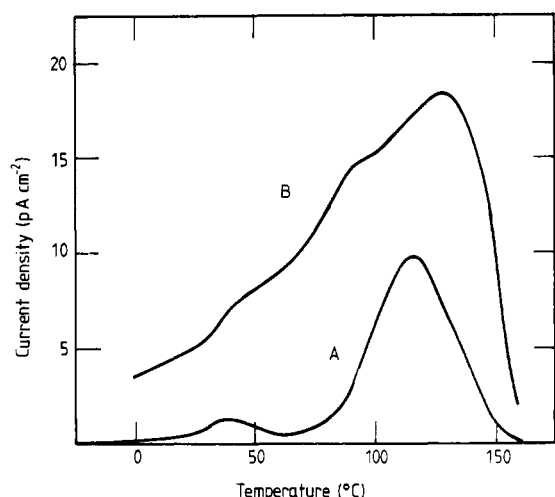


Figure 9. Short-circuit TSD thermograms. Curves: A, sample with Al electrodes, after polarising with 2 MV cm^{-1} at 80°C for 30 min; B, after bombardment with 10 keV electrons in short circuit, dose $5 \mu\text{C cm}^{-2}$, at 20°C . Ramp rate = $4.5^\circ\text{C min}^{-1}$.

grams on voltage-polarised samples with Au electrodes resembled curve A when polarised at low voltage but resembled curve B when polarised at high voltage.

3.5. Electron-beam depolarisation experiments

The results of the EBD measurements on films voltage polarised at RT or 100°C were close to the limits of resolution of the method, and are mostly qualitative. They may be summarised as follows: the net space charge detected below an Al anode was very small but unmistakably negative, i.e. of heterocharge character. The term 'net' indicates that the experiment cannot resolve heterocharge from homocharge if these coexist. The net charge below an Al cathode was negative (homocharge). The results for an anode of Au were inconclusive, but if significant hole charge had been injected, this could not have escaped detection. Surprisingly, however, a cathode of Au appeared to inject electrons more efficiently than one of Al. At RT, the injected charge was still too weak to be quantified, but at 100°C it amounted to a few per cent of CV (C = capacitance, V = polarising voltage).

4. Discussion

4.1. Dielectric loss

We summarise the behaviour of the transients: $I_r(t)$ always decays uniformly as t^{-1} . At and below RT, $I_a(t)$ behaves similarly but, above RT, hyperbolic decay is observed only initially (§ 3.1). In the hyperbolic decay regime of $I_a(t)$, the isochronal volume current is ohmic (§ 3.3). We follow the usual interpretation of such behaviour in terms of dipolar relaxation with

broadly distributed relaxation times (e.g. Jonscher 1983), as was done by Das Gupta and Joyner (1976).

The dielectric loss factor $\tan \delta$ may be deduced from the $I_r(t)$ transients which are unperturbed by conduction. Using the expression (Baird 1968)

$$\tan \delta \approx 1.6 t I_r(t) / CV \quad (3)$$

we calculate $\tan \delta$ as 0.6×10^{-4} at RT, 1.2×10^{-4} at 100°C , and 0.3×10^{-4} at -175°C . Since the product $t I_r(t)$ is invariant with t within the limits of resolution of our experiments, $\tan \delta$ appears to be constant over the entire time span (10^{-5} to 10^3 s) which, in the frequency domain, corresponds to 10^4 to 10^{-4} Hz (Hamon 1952). The loss at RT is an order of magnitude lower than that similarly deduced by Das Gupta and Joyner (1976) from current transients, or by Anderson and McCall (1958) from AC measurements. It is, however, in order-of-magnitude agreement with the loss measured by Buckingham and Reddish (1967) for their lowest-loss PP, which our material resembles. An attempt to measure the loss of our samples on a frequency analyser (by courtesy of J Pugh, KQC College, London) proved inconclusive on account of insufficient resolution of this instrument.

Garton (1946) and Atkinson and Fleming (1980b) have already pointed out that dielectric loss measurements are influenced by surface conduction. If we calculate a volume-loss factor $\tan \delta_v$ by multiplying the right-hand side of equation (3) by v_r , we now obtain a weakly structured loss spectrum at RT (see figure 10), which reflects the time (or frequency) dependence of v_r . The curve is for 0.5 MV cm^{-1} , but should remain unchanged as $F \rightarrow 0$ (cf figure 5(a)). The weak loss peak at $f \approx 3 \text{ kHz}$ probably

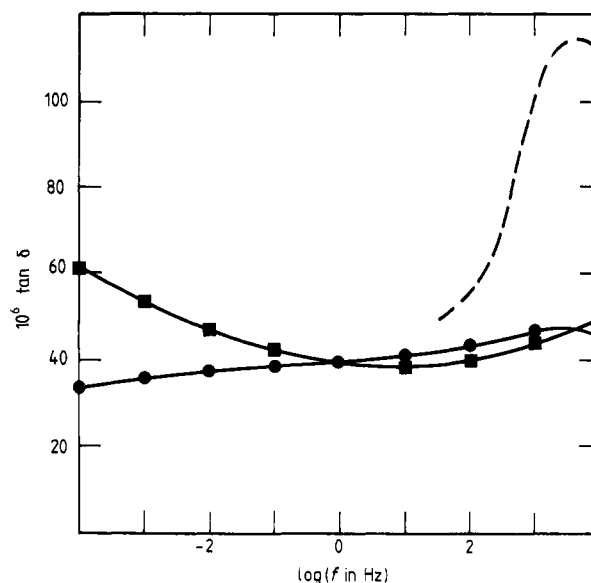


Figure 10. The dispersion of the dielectric loss factor $\tan \delta$. ●, volume loss at 20°C ; ■, volume loss at 100°C ; both at 0.5 MV cm^{-1} , calculated from the volume component of resorption current. Broken curve: AC measurements by Buckingham and Reddish (1967) at 20°C .

corresponds to the β -relaxation peak found at ≈ 5 kHz by Buckingham and Reddish (1967) in isotactic PP (broken curve in figure 10), and at ≈ 10 kHz by Work *et al* (1964) in atactic PP. Our peak is much less intense than that of Buckingham and Reddish but, on the other hand, the AC measurements represent the sum of volume and surface losses, that is they depend on the electrode size employed, and thus cannot be taken as definitive.

The dispersion curve for 100 °C in figure 10 is also for 0.5 MV cm⁻¹, and should remain unchanged as $F \rightarrow 0$ (cf figure 6(b)). Evidently, the β -peak has moved out of range to higher frequencies, but there is a suggestion of a new relaxation at very low frequencies.

The fact that the broad, weak, peak centred at ≈ -70 °C in figure 3 appears to be stationary with sample time (i.e. frequency) implies that the associated activation energy is very low. This peak probably corresponds to one of similar properties observed by Das Gupta and Joyner (1976) at approximately -50 °C. We are unable to relate it to any known loss mechanism in PP. Nor can we test whether it is in any way connected with surface leakage, as we have not determined ν and s below RT.

4.2. Conduction mechanism

In contrast to its behaviour observed by us in PET (§ 1), in PP the absorption tail is insensitive to contact metal; its profile at 2 MV cm⁻¹ closely resembles that at 0.5 MV cm⁻¹; it shows no tendency to level out; its isochronal volume component is only slightly supraohmic; and its presence does not perturb the uniformly hyperbolic decay of $I_r(t)$. These observations all point to the absence of any significant build-up of injected electronic charge, which otherwise might be responsible for the conduction tail.

The activation energy of approximately 0.8 eV for the conduction process (§ 3.1) is in approximate agreement with the value found by Das Gupta and Joyner (1976). These authors suggested that the tail might be due to bulk electronic conductivity, with a low hopping mobility and blocking contact conditions. Irrespective of any detailed considerations of the origin of the carriers and of the transport mechanism (which involve conceptual difficulties), however, the apparently total absence of tail current at RT (figure 2), and its abrupt onset just above RT (figure 3), cannot be reconciled easily with a conduction mechanism involving only the single activation energy of 0.8 eV.

The same objection applies to ionic conductivity as proposed by Foss and Dannhauser (1963), unless an additional assumption is made. We suggest that, below the glass transition temperature, T_g (≈ -8 °C), the ions are firmly trapped in the amorphous regions of the polymer. As the temperature is increased and the structure relaxes, they become mobile until fully 'free', subject to their diffusion activation energy ΔE ,

at approximately 25 °C. We propose that the release from traps is described by the WLF expression (Williams *et al* 1955) for the relaxation of the amorphous phase at T_g . The ionic conductivity σ for $T > T_g$ would then be of the form

$$\sigma \propto \exp(-\Delta E/kT) \{1 + C \exp[-0.8 T'/(1 + 0.02 T')]\}^{-1} \quad (4)$$

where $T' = T - T_g$, C is a trapping parameter, and the usual values are used for the numerical parameters in the WLF term. For example, if $C = 10^3$, the denominator would drop from 10^3 to 1 between -8 and 25 °C. An analogous model for electronic conductivity, with a trapping threshold related to T_g , would be implausible.

We therefore agree with Foss and Dannhauser's interpretation of the conduction as ionic. This is compatible with the only weakly supraohmic field dependence of I_a , and the absence of any perturbation in I_r , since the internal polarisation field will remain small. However, our value of 0.8 eV for the ionic activation energy is far below that of 1.5 eV quoted by Foss and Dannhauser (1963). This suggests that the ions are impurities, which are likely to differ in samples of different manufacture.

4.3. TSD experiments

The most prominent feature of the thermograms of voltage-polarised samples with Al electrodes (figure 9 (curve A)) is the peak at 115 °C (we comment on the effect of Au electrodes in § 4.4). Matsui and Murasaki (1973) observed a main peak lying between 75 and 95 °C in voltage-polarised specimens with Cu electrodes. This peak could correspond to ours, but Baba and Ikezaki (1982) and Keller and Datt (1985), in corona-charged films, recorded peaks at temperatures close to 115 °C. The latter authors deduced an activation energy of ≈ 1 eV which is comparable with our value of 0.85 eV.

The theory of dipolar TSD (Bucci *et al* 1966) predicts for the temperature T_m of a peak representing a Debye-like relaxation

$$T_m^2 = 0.16 bE/kf(T_m) \quad (5)$$

where b is the heating rate, E the activation energy of the relaxation, and $f(T_m)$ the frequency at which the same relaxation would produce a loss peak in an isothermal AC measurement at T_m . (Van Turnhout (1975) derives a factor 0.113 instead of 0.16.) For $T_m = 388$ K, we calculate $f(T_m)$ to be approximately 1 mHz. If $f(T)$ obeyed a simple Arrhenius law $f(T) = \text{constant} \exp(-E/kT)$, we could estimate the frequency at which the loss would occur at RT, and find $f(295) \approx 0.3$ μ Hz. However oversimplified the assumptions made, it is clear that the 115 °C peak cannot possibly be associated with the β -relaxation (≈ 5 kHz at RT).

On the other hand, the similar activation energies

of the 115 °C peak (0.85 eV) and of the conduction tail (0.8 eV) suggests associating the peak with the ionic conduction mechanism assumed. At 100 °C, the corresponding AC loss peak would be expected near 0.3 mHz. In figure 10, the loss at 100 °C must peak below 0.1 mHz, but there is at least agreement in principle. It is possible that the TSD peak at 35 °C represents a second ion species.

The short-circuit thermogram after electron bombardment (figure 9(curve B)) closely resembles that recorded by Ieda *et al* (1980) under an external field after x-irradiation. Both methods of excitation leave trapped electrons and holes which have escaped recombination, but under electron bombardment the electron charge is the greater and provides the internal field which drives the TSD current. Without an external field, this current must cease when the electron traps are discharged. We therefore interpret the peak at 130 °C, close to the softening point of the polymer, as representing electron traps of depth ≥ 0.9 eV, located in the crystalline phase.

Baba and Ikezaki (1982) show a peak at 70 °C after positive corona charging. Since this is likely to inject holes, we are tempted to associate the hump at ≈ 80 °C in figure 9(curve B) and in the thermogram of Ieda *et al* (1980) with hole traps of estimated depth $25 kT_m \approx 0.75$ eV. We attribute the background shoulder in figure 9(curve B) and in the thermogram of Ieda *et al* (1980) to the ionic conductivity. This would explain the otherwise puzzling observation of a finite depolarisation current at 0 °C.

4.4. EBD experiments

The important observation of heterocharge, at least at an Al anode (§ 3.5), supports the assumption of ionic conductivity even at RT. The detection of electron injection does not contradict the conclusion that such injection makes no contribution to the tail current, since trapped electrons are mobilised only near the softening temperature.

The observation of stronger electron injection by a cathode of Au than by one of Al, however, is perverse, since Au possesses the higher work function. We dismiss this result as an experimental artefact. Au is evaporated at a higher temperature than Al. While aluminised films look perfectly smooth, Au-coated films appear slightly wrinkled. We suggest that the Au/PP interface contains asperities or 'hot spots', at which the local field, during polarisation, is much higher than the average field, and promotes field emission. The injected electrons are deeply trapped at the end of their range, and produce a local back-field which prevents further field emission. The emission, therefore, would take place only in short, non-destructive, bursts, but the trapped electrons would be detected on depolarisation. The presence of this space charge presumably accounts for the resemblance of thermograms of Au-electroded samples,

after strong voltage polarisation, to those recorded after electron bombardment (cf § 3.4).

4.5. Surface leakage

In our sample geometry, leakage from the live to the counter electrode around the edge of the film would be intercepted by the guard ring and not be detected. The surface current recorded, therefore, flows in or out of the counter electrode across the annular space between it and the guard ring. Since the width of this annulus is always much greater than the film thickness, the fringe field at the electrode edge and its lateral component are not screened out by the guard ring (Garton 1946).

Several attempts have been made to explain the time and field dependence of the surface current, mostly in terms of the spread of electronic charge injected by the metal edge. In this case, the profile of the edge is important (Lachish and Steinberger 1974), but this was not deliberately altered in our experiments, nor were Au electrodes investigated. The present state of the theory has been reviewed by Lowell (1982, 1983), who concludes that it is still unsatisfactory. Certainly, our results (figures 4–7) cannot be reconciled with any existing model.

In the measurements with Al electrodes (§ 3.2), no differences were found when the polarity was reversed. This argues against electronic injection. Atkinson and Fleming (1976) suggest that the surface current may be provided by electrons in occupied surface states. However, in view of the presence of ionic bulk conductivity in PP, it seems likely that the surface current also contains an ionic in addition to a dipolar component.

5. Conclusion

The results of the transient and depolarisation experiments on PP are so different from our earlier results on PET, that we are led to the firm conclusion that contact-injected electronic charge cannot account for the conduction observed in PP at elevated temperatures. On the other hand, the observations are all compatible with an ionic conduction mechanism which comes into play above the glass transition temperature. However, neither the sign nor the nature of the predominantly mobile ion species could be determined. There is little doubt that the ions are of impurity origin. Weak electron injection was detected, but since there appear to be electron traps so deep that they release electrons only just below the softening temperature of the polymer, the injected electrons would be expected to become deeply trapped close to the cathode.

The dependence on time, field and temperature of the surface contribution to the transient current is highly complex. The surface current may itself be the sum of dipolar and ionic components, but no attempt

was made to model this. It is of interest to note that, without correcting for the surface component, the field dependence of the bulk current could not have been correctly determined. We also note that the transient records, despite the degree of precision which we thought to have achieved, are of insufficient resolution to reproduce the dispersion of dielectric loss by themselves, although some structure becomes apparent in the resolved volume loss spectra.

Acknowledgments

We wish to thank S L Miller for pointing out that the complexity of the surface current evolution suggests a two-component process, and to acknowledge financial support from the SERC (UK).

References

- Anderson E W and McCall D W 1958 *J. Polym. Sci.* **31** 241
- Atkinson P J and Fleming R J 1976 *J. Phys. D: Appl. Phys.* **9** 2027
- 1980a *J. Phys. D: Appl. Phys.* **13** 625
- 1980b *J. Phys. D: Appl. Phys.* **13** 655
- Baba A and Ikezaki K 1982 *Appl. Phys. Lett.* **40** 1027
- Baird M E 1968 *Rev. Mod. Phys.* **40** 219
- Bucci C, Fieschi R and Guidi G 1966 *Phys. Rev.* **148** 816
- Buckingham K A and Reddish W 1967 *Proc. IEE* **114** 1810
- Das Gupta D K and Joyner K 1976 *J. Phys. D: Appl. Phys.* **9** 2041
- Foss R A and Dannhauser W 1963 *J. Appl. Polym. Sci.* **7** 1015
- Garton C G 1946 *J. Inst. Electr. Eng.* **93** 397
- Grossweiner L I 1953 *J. Appl. Phys.* **24** 1306
- Hamon B V 1952 *Proc. IEE* **99** 151
- Hirsch J, Ko A Y and Irfan A Y 1984 *IEEE Trans. Electr. Insul.* **EI-19** 190
- Ieda M, Mizutani T and Suzuoki Y 1980 *Mem. Fac. Eng. Nagoya Univ.* **32** 173
- Jonscher A K 1983 *Dielectric Relaxation in Solids* (London: Chelsea Dielectrics)
- Keller J M and Datt S C 1985 *Phys. Status Solidi a* **91** 205
- Ko A Y-Y 1984 *PhD Thesis* University of London
- Ko A Y and Hirsch J 1981 *Solid State Commun.* **39** 215
- Kurtz S R and Anderson R A 1986 *J. Appl. Phys.* **60** 681
- Lachish U and Steinberger I T 1974 *J. Phys. D: Appl. Phys.* **7** 58
- Lowell J 1982 *J. Phys. D: Appl. Phys.* **15** 109
- 1983 *J. Phys. D: Appl. Phys.* **16** 2223
- Matsui M and Murasaki N 1973 *Electrets* ed. M M Perlman (Princeton, NJ: Electrochemical Society) p 141
- Milnes A G 1973 *Deep Impurities in Semiconductors* (New York: Wiley)
- Quynn G G, Riley J, Young D A and Noether H D 1959 *J. Appl. Polym. Sci.* **2** 166
- Tavakoli-Juibari M B 1987 *PhD Thesis* University of London
- Tavakoli M and Hirsch J 1987 *Solid State Commun.* **63** 25
- Van Turnhout J 1975 *Thermally Stimulated Discharge of Polymer Electrets* (Amsterdam: Elsevier)
- Williams M L, Landel R F and Ferry J D 1955 *J. Am. Chem. Soc.* **77** 3701
- Work R N, McCammon R D and Saba R G 1964 *J. Chem. Phys.* **41** 2950